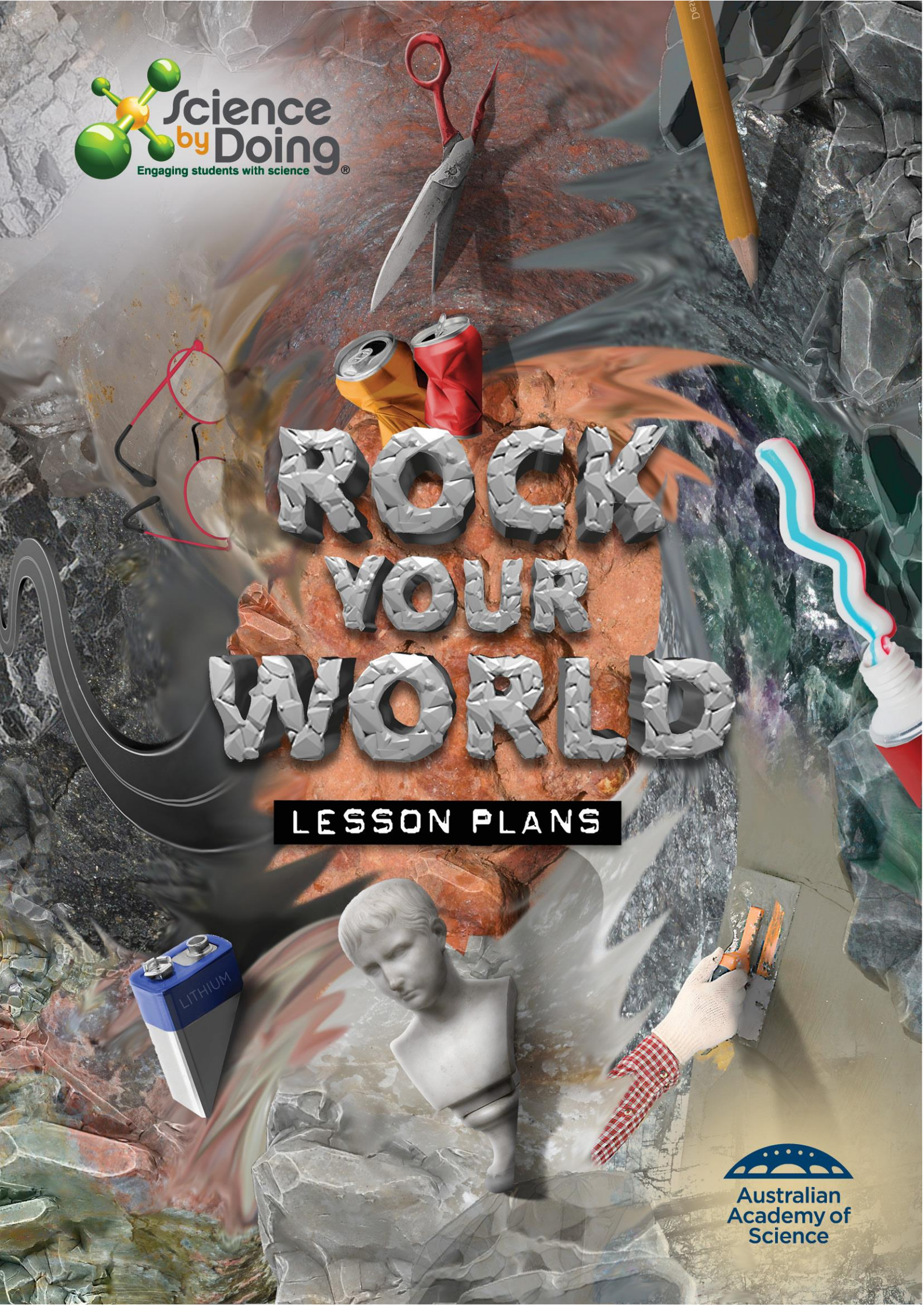




ROCK YOUR WORLD

LESSON PLANS

Part 1 Rocks – part of the Earth system

The core activities in this unit total about 22 hours of lesson time. The Content Descriptions (Australian Curriculum: Science) for each activity are listed here as curriculum codes and described fully in each lesson plan.

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
1.1	Every rock tells a story	ACSSU153 ACSHE 134 AC SIS148	Engage & Explore	In an animation Connie Conglomerate introduces some of the rocky characters, and they each tell a little about their history.	Core
			Digital		
			Short		
1.2	The deep Earth	ACSSU153 ACSHE 134 ACSHE136 AC SIS148	Engage & Explore	Introduces the inner structure of the Earth.	Core
			Digital		
			Short		
1.3	Cycling rocks	ACSSU153 AC SIS145 AC SIS148	Explore & Explain	A first introduction to the idea that rocks change and are recycled through Earth's processes.	Core
			Digital		
			Short		
1.4	It's all in the crust	ACSSU153 AC SIS144 AC SIS145 AC SIS148	Explore & Explain	Students examine and describe basic rock types (sedimentary, igneous and metamorphic) according to their characteristics and relate them to their processes of formation.	Core
			Hands-on & Digital		
			Medium		

Part 2 Minerals – ingredients in the rock recipe

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
2.1	Rock or mineral?	ACSSU153 ACSIS145 ACSIS148	Explore & Explain Hands-on & Digital Short	Students look closely at some rocks to discover their mineral composition and determine the difference between rocks and minerals.	Core
2.2	Diagnostic properties	ACSSU153 ACSIS145 ACSIS148	Explore & Explain Hands-on & Digital Medium	Students examine various minerals and determine diagnostic properties of minerals that can be used to tell minerals apart.	Optional
2.3	Testing minerals	ACSSU153 ACSIS145 ACSIS234 ACSIS148	Engage, Explore, Explain & Elaborate Hands-on Medium	Students are introduced to mineral tests and use them to differentiate between biotite and quartz. Students then use the tests to identify two unknown mineral samples.	Core
2.4	Rock identification	ACSSU153 ACSIS145 ACSIS148	Explain & Elaborate Digital Short	Students identify the minerals in three rock specimens and then use this information to identify the three rocks.	Core
2.5	Mining our everyday minerals	ACSSU153 ACSHE 134 ACSHE 135 ACSHE136 ACSIS148	Explore Digital Short	Students learn how important minerals are in producing many of our everyday objects and learn to associate various minerals with their uses. Students study Australia's export industries and appreciate the large role minerals and fossil fuels make to our economy.	Core

Part 2 Minerals – ingredients in the rock recipe (continued)

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
2.6	Careers in mineral exploration	ACSSU153 ACSHE 134 ACSHE 135 ACSHE136 ACSIS148	Explore & Explain	Students explore the challenges of extracting valuable ores and minerals from their parent rocks. Students explore the various career pathways in the mining industry and learn where deposits and mines are located in Australia.	Core
			Digital		
			Long		
2.7	Money for rocks	ACSHE136 ACSIS148	Explain & Elaborate	Students are presented with five minerals and asked which contributes most to Australia's economy.	Optional
			Digital		
			Short		

Part 3 Rocks from the Earth's Furnace – igneous rocks

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
3.1	Big and small crystals	ACSSU153 ACSIS140 ACSIS145 ACSIS234 ACSIS148	Engage, Explore, Explain & Elaborate	Students investigate the effect of temperature on the formation of salol crystals and relate their experience to the formation of igneous rocks.	Core
			Hands-On		
			Medium		
3.2	Erupting and intruding	ACSSU153 ACSIS148	Explore, Explain & Evaluate	Students learn about the distinction between volcanic and plutonic rocks. They also learn the terms 'extrusive' and 'intrusive'.	Core
			Hands-on & Digital		
			Medium		

Part 4 Grain by grain – sedimentary rocks

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
4.1	Breaking rocks	ACSSU153 ACSIS139 ACSIS140 ACSIS141 ACSIS144 ACSIS145 ACSIS146 ACSIS234 ACSIS148	Engage, Explore & Explain Hands-on & Digital Long	Students simulate the processes of physical and chemical weathering using sugar cubes. They further explore weathering and erosion in a video.	Core
4.2	Which sediments where?	ACSSU153 ACSIS145 ACSIS148	Explore, Explain & Elaborate Digital Short	Students explore different sedimentary environments and relate this to their own region.	Core
4.3	Sticking things together	ACSSU153 ACSIS139 ACSIS140 ACSIS141 ACSIS144 ACSIS145 ACSIS146 ACSIS234 ACSIS148	Engage, Explore & Explain Hands-on Long	Students make various rocks from a range of sediments and binders. Students design fair tests to determine which rocks are harder and more durable. Students record their results in both a table and a graph and draw conclusions from their results.	Core
4.4	Grains and rocks	ACSSU153 ACSIS145 ACSIS146 ACSIS234 ACSIS148	Explore & Explain Hands-on & Digital Medium	Students examine three different sedimentary rocks and identify their key characteristics.	Core
4.5	Making rocks from body parts	ACSSU153 ACSHE 135 ACSIS141 ACSIS144	Explore & Explain Hands-on & Digital Medium	Students explore the formation and characteristics of some biogenic rocks, including limestone and coal.	Core

Part 5 Baking and burying: metamorphic rocks

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
5.1	These rocks are so hot	ACSSU153	Engage, Explore & Explain	Students bake modelling clay and compare its characteristics with unbaked clay. They relate this to the formation of some metamorphic rocks.	Core
		ACSIS139			
		ACSIS140			
		ACSIS141	Hands-on & Digital		
5.2	Regional metamorphosis	ACSIS144		Students read about the famous tilted beds of metamorphic rocks in northern Scotland that extend for about 250 km and see the scale on which burial can occur; leading to large scale regional metamorphism. Students conduct a simple acid fizz test to identify that limestone and marble have similar chemical properties.	Core
		ACSIS145			
		ACSIS234	Hands-on & Digital		
		ACSIS148	Short		
5.3	A tortured history	ACSSU153	Explain & Elaborate	This activity summarises the ideas of contact metamorphism and regional metamorphism into a simple diagram that illustrates which rocks form due to increased temperature or pressure, or both. Students learn about the life of Neville Gneiss and watch two more short videos that explore metamorphic rocks.	Core
		ACSHE136			
		ACSIS148	Digital		
			Short		

Part 6 The rock cycle and geological time

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
6.1	Rocks into rocks	ACSSU153 ACSHE226 ACSHE136 ACSIS146 ACSIS148	Engage, Explore, Explain, Elaborate & Evaluate Hands-on & Digital Long	Students consolidate their understanding through activities exploring the rock cycle. These include an on-line interactive quiz and a hands-on activity modelling different aspects of the rock cycle.	Core
6.2	On the level	ACSSU153 ACSIS144 ACSIS145 ACSIS146 ACSIS234 ACSIS148	Explore & Explain Hands-on Medium	This is a hands-on activity where students simulate the deposition of sediments to investigate: - the relative ages of sediments, - the horizontal orientation of the beds and - the nature of graded bedding.	Optional
6.3	How do fossils form?	ACSSU153 ACSHE 134 ACSIS146 ACSIS234 ACSIS148	Engage, Explore & Explain Digital Short	Students complete an interactive exercise illustrating the formation of fossils. In the Notebook exercise students learn about South Australia's world heritage listed Naracoorte Caves and the fossils found there of Australia's ancient megafauna. Students watch a video (6 minutes) on palaeontology in Australia which introduces stratigraphy (which is further developed in Activity 6.4). In the Find out more section students may explore the range of fossils found in Australia by exploring links to the Queensland and Melbourne museums.	Core
6.4	Is this the same as that?	ACSSU153 ACSIS146 ACSIS234 ACSIS148	Explore, Explain & Elaborate Classroom Medium	Students work in small groups to sequence cards that represent different rock layers containing fossils. Sequencing the rock layers simulates how palaeontologists use fossils to achieve stratigraphic correlation.	Core

Part 6 The rock cycle and geological time (continued)

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
6.5	Relative geological time	ACSSU153 ACSIS148	Explore, Explain & Elaborate	Students explore the horizontal bedding and sheer majesty of the Grand Canyon. Using age data for each of the layers, they map onto a diagram of the Grand Canyon the name of the geological period when the layer was deposited and possible fossils that could be found from that period.	Core
			Digital & Classroom		
			Medium		
6.6	How long does it take?	ACSSU153 ACSIS145 ACSIS146 ACSIS234 ACSIS148	Explore, Explain & Elaborate	Students roll dice in the Rock Cycle Game to follow their rock through the rock cycle and experience the magnitude of geological time. This activity links students' understanding of the Rock Cycle with their understanding of the Relative Geological Time Scale.	Core
			Digital & Classroom		
			Medium		
6.7	Forming landforms	ACSSU153	Explain & Elaborate	Students investigate several Australian landforms and the different environmental processes that led to their formation.	Core
			Digital		
			Short		
6.8	Sample test		Evaluate	A sample summative test (editable version) and marking scheme are available to teachers by email request from <i>Science by Doing</i> (sbd@science.org.au). In your email request please include your school (or university) and contact details. Note - <i>Science by Doing</i> provides sample assessment items and whilst every effort has been made, the security of these items cannot be guaranteed. <i>Science by Doing</i> encourages teachers to modify the items to suit individual teaching programs.	Optional
			Classroom		
			Long		

1.1 Every rock tells a story

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Rocks are more than just small features that are found occasionally lying on the ground. They are the fundamental part of the crust on which we live.
 - There are different types of rock, which form under different conditions and have different geological histories.
 - Although most rocks appear to be endlessly durable, they have a life story. Rocks are broken down and rebuilt over very long periods of time.
 - All rocks on Earth can be broadly categorized into three groups; sedimentary, igneous and metamorphic. These groups of rocks have different characteristics that relate to the manner in which they are formed.
- ACSHE 134 Scientific knowledge has changed peoples' understanding of the world and is refined as new evidence becomes available.
- ACSI148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- describe some different rock types
- identify that different rocks form under different conditions and have different histories.

What ideas might your students already have?

- students may not identify rocks as natural materials.
- students will not be familiar with the ideas that rocks can form under different conditions.
- students may conceptualise 'rocks' in different ways, involving particular sizes and shapes.

Key vocabulary:

Rock, sedimentary, igneous, sandstone, conglomerate, basalt.

Equipment list

The CLASS will require:

- Internet access

Each STUDENT will require:

- **Notebook**

Things to consider

For hearing impaired students you may provide the video transcript.



The idea that identification of rocks and minerals relates to the processes that formed them, and potentially tells us about systems that existed in the past, makes this whole subject area a great deal more interesting than simply being able to distinguish sandstone from granite. If we understand why we might want to identify rocks and minerals in different places, for example to understand the environment of deposition of a sediment (i.e. sediments formed on a reef are very different from sediments deposited by a glacier) it makes learning their properties more relevant to the students. Using this lens also allows us to address the **Science as a Human Endeavour** aims for this subject area.

Learning about a wide range of rocks and minerals so we can then competently teach this to students is a little intimidating. The good news is that if we (and the students) learn the most common rock types, and about a dozen or so minerals, we are well placed to identify almost every rock that we are likely to encounter in our lives. This is because there are only a finite number of processes that form rocks, and there are a relatively small number of so called 'common rock forming minerals' that make up most rocks.

The construction of the *Science by Doing* on-line resource, *Rocks and Minerals*, linked to the *Rock Your World* unit provides information on the most common rocks and minerals so that we can learn the essence of this subject area readily. In the first instance this learning can take place in parallel with the students, and then we will become more confident each time this subject area is taught.

The *Rocks and Minerals* collection has been compiled from high quality photographs of rocks and minerals that mostly appear as we are likely to see them in the field. For most of these images we have not used perfect museum-quality specimens, but preferentially selected examples that illustrate the properties we would normally see if we found a really good example of that rock or mineral type in a road cutting.

The idea is that the students will know the key rocks and minerals well. The rocks and minerals that do not appear in their local area can be examined using the school collection or the online collection *Rocks and Minerals*.

Another way to use the *Rocks and Minerals* collection is to sort your own school collections according to the structure provided here. This will mean that you only have about 20 rocks and 10 to 15 minerals to learn for core teaching, making the task less daunting for you and the students.

Ideally you would have multiple specimens of the same rock and mineral types so that you have constructed a class set. If you have spectacular specimens, use these for display purposes, so you keep the teaching focus on the common rocks and minerals.

A safety alert was issued regarding the potential for school rock collections to have samples of asbestos. Ensure that you have checked that your collections do not contain any specimens containing asbestos or any ores that may be radioactive.

Teacher content information

Students may not have considered that the Earth is made of rock material right to the core.



- ▲ As a teacher, it is your responsibility to follow your school safety and risk, policies and procedures.
- ▲ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

- ▲ Always ensure safe and ethical disposal of chemicals.
- ▲ Use teacher judgement when considering equipment substitutions.
- ▲ Make considered choices selecting and using resources to minimise waste.

Commonly, in junior primary school, they may have cut open an orange to illustrate that the lithosphere or Earth's crust is a very small thickness on the outer part of the Earth. There is a perception that the lithosphere is entirely made up of the rocks that we observe in our local area, but we need to make it clear that there is a much broader range of rock types present around the Earth and throughout the lithosphere depending on the geological processes that have taken place. This brings in a little geography and 3D thinking. Limited time is likely to have been spent on the rocks that make up the deeper parts of the Earth, but the foundation concepts presented here will help explain this in subsequent years.

Teachers may also refer to the Year 9 unit *Big Systems* to learn more about the detailed structure of the interior of the Earth and how scientists have derived this information.

Key idea

Rocks are more than just small features that are found occasionally lying on the ground. They are the fundamental part of the crust on which we live. There are different types of rock, which form under different conditions and have different geological histories.

Lesson plan

Step 1: Begin the lesson by showing a rock and posing these questions to the class:-

How old are rocks?

How are some rocks different to others?

What are some of the different conditions that rocks may experience as they form?

Step 2: Introduce students to **Activity 1.1** in the *Student Guide*.

Step 3: Show the video to the class. You could project the video onto a whiteboard.

Step 4: The **Notebook** activity could be completed as a whole class discussion or individually.



▲ As a teacher, it is your responsibility to follow your school safety and risk, policies and procedures.

▲ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

▲ Always ensure safe and ethical disposal of chemicals.

▲ Use teacher judgement when considering equipment substitutions.

▲ Make considered choices selecting and using resources to minimise waste.

1.2 The deep Earth

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- The Earth consists of several distinct layers, including the crust, mantle, outer core, and inner core. Conditions within the mantle and core are very different to those at the surface. A combination of high temperatures and pressures produce conditions that are quite different to those found at the surface.
- ACSHE 134 Scientific knowledge has changed peoples' understanding of the world and is refined as new evidence becomes available.
- ACSHE136 People use science understanding and skills in their occupations and these have influenced the development of practices in areas of human activity.
- AC SIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- describe that conditions deep within the Earth are hotter and experience higher pressures than at the surface.

What ideas might your students already have?

students know that some materials come from deep mines, but will not be aware of the conditions experienced in these mines.

Key vocabulary:

Crust, mantle, outer core, inner core.

Equipment list

The CLASS will require:

- internet access

Each STUDENT will require:

- Notebook**

Things to consider

For hearing impaired students you may provide the video transcript.

There are some simple ways of illustrating these concepts like the difference between lying under one blanket and lying under 20 blankets or 100 blankets. The student would feel more pressure (load) and more heat (due to insulation by the blankets) in the later scenarios. If you continue to use the same analogy, now have the student consider how they would feel if the blankets were progressively removed. They would have more freedom to move (expand) and they would feel cooler. If we remove layers of rock by erosion, the underlying rocks can 'relax' (expand). For example, the joints in granites might progressively open as the granite becomes exposed at the surface.



Hot magma will cool more rapidly as it gets closer to the Earth surface, and eventually quench when it erupts at the Earth surface, because there is a greater temperature contrast between the magma and the enclosing medium (e.g. surrounding rock, air).

Teacher content information

Refer to the notes in **Activity 1.1** of the *Teacher Guide*.

Key Idea

The Earth consists of several distinct layers, including the crust, mantle, outer core, and inner core. Conditions within the mantle and core are very different to those at the surface. A combination of high temperatures and pressures produce conditions that are quite different to those found at the surface. As we approach the surface the conditions progressively change (the idea of a gradient) until they reach ambient conditions at the Earth surface.

Lesson plan

Step 1: Ask students what they think it is like deep inside the Earth. How can we learn what it is like inside the Earth?

Step 2: Introduce students to the *Student Guide* and present the two videos.

Step 3: The **Notebook** activity could be completed as a whole class discussion or individually.

Suggested Questions

Lava flowing out of some volcanoes has a liquid consistency, but rocks deep within the Earth, in the mantle, are not completely liquid. How do geologists use the concepts of heat and pressure to explain this?



1.3 Cycling rocks

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Although most rocks appear to be endlessly durable they have a life story. Rocks are broken down and rebuilt over very long periods of time.
 - Rock materials are not destroyed but cycle through the processes of formation of sedimentary, igneous and metamorphic rocks, over millions of years.
- ACSIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.
- ACSIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- describe simple processes involved in the rock cycle
- explain that rocks are formed and broken down continually

What ideas might your students already have?

- many students feel that rocks are permanent features, and have few concepts related to the formation and breaking down of rocks.

Key vocabulary:

Sediment, sedimentary, igneous, lava, magma

Equipment list

Each STUDENT will require:

- internet access
- **Notebook**

Things to consider

This activity provides students with a brief introduction to the rock cycle. A more detailed treatment of this topic occurs in **Activity 6.1**. The idea of introducing this near the beginning of the unit serves two main purposes. It reinforces the idea that understanding rocks relates to learning the processes that formed them, and it allows us to revisit the rock cycle later in the program of work in a more comprehensive way.

In this simple treatment of the rock cycle the concept of metamorphism is not included. This is dealt with in some detail in **Part 5**.

Teacher content information

Essentially the rock cycle reinforces the concept that matter is neither created nor destroyed but is transformed into different configurations. This is very big idea in science.



In summary:

If any rock materials are weathered and eroded they will form sediment, which can be cemented and compacted to become a (clastic) sedimentary rock. Other sedimentary rocks form either by precipitation from a supersaturated solution (chemical), or due to the accumulation of plant or animal matter (biogenic).

If any rock materials are subjected to sufficient heat, the composite minerals melt and magma (hot molten rock) forms. When this magma cools it forms igneous rocks. If magma cools slowly below the surface there is time for larger crystals to form and the rocks are called plutonic rocks (e.g. granite). Magma may erupt at the surface to violently eject block and ash material (volcanic sediments) or it may erupt to produce lava flows which cool quickly to form fine-grained volcanic rocks (e.g. basalt).

The other concept that students (and most adults) struggle to truly comprehend, is geological time. The span of geological time is immense, however, if we focus on that component of the geological record that is relevant to the processes we are learning about in a particular area, this makes working with geological time a little more manageable.

Key Idea

Although most rocks appear to be endlessly durable they have a life story. Rocks are broken down and rebuilt over very long periods of time.

Lesson plan

Step 1: Introduce students to the *Student Guide* and simplified rock cycle.

Step 2: The **Notebook** activity could be completed as a whole class discussion or individually.



1.4 It's all in the crust

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- The Earth consists of several distinct layers, including the crust, mantle, outer core, and inner core. Conditions within the mantle and core are very different to those at the surface. A combination of high temperatures and pressures produce conditions that are quite different to those found at the surface.
 - All rocks on Earth can be broadly categorized into three groups; sedimentary, igneous and metamorphic. These groups of rocks have different characteristics that relate to the manner in which they are formed.
- ACSIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.
- ACSIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence..
- ACSIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- describe key features of sedimentary, igneous, and metamorphic rocks
- apply key characteristics to identify some rock samples
- explain how the characteristics of different rock types relate to the manner in which they were formed.

Key vocabulary

Metamorphic, sedimentary, igneous

Equipment list

Each GROUP will require:

- collection of rocks including common examples of the 3 rock groups
- dissecting microscope
- camera (if available)
- internet access

Each STUDENT will require:

- Notebook



Things to consider

In the notes for **Activity 1.1** we discussed the idea of arranging the school rock collection in a manner similar to that illustrated by the **Rock your World** online *Rocks and Minerals* collection. The students could help with this by identifying and sorting (curating) components of the collection as part of an activity.

Are your students familiar with the use of dissecting microscopes?

A safety alert was issued regarding the potential for school rock collections to have samples of asbestos. Ensure that you have checked that your collections do not contain any specimens containing asbestos or any ores that may be radioactive.

Teacher content information

Refer to the notes on sedimentary and igneous rocks in **Activity 1.3** of the *Teacher Guide*.

Metamorphic rocks have been changed over time by extreme pressure and heat (over 150°C). These rocks can be formed by the high pressures deep under the Earth's surface and from the extreme heat caused by magma. Uplift and erosion help bring metamorphic rock to the Earth's surface.

Examples of metamorphic rocks include anthracite (from coal), quartzite (from sandstone), marble (from limestone) and slate (from shale or mudstone).

Key Idea

All rocks on Earth can be broadly categorised into three groups; sedimentary, igneous and metamorphic. These groups of rocks have different characteristics that relate to the manner in which they are formed.

Lesson plan

Step 1: Form working groups of two to four students. Support students in discerning the differences between the three rock types. Assist students to use the dissecting microscopes to see more detail.

Step 2: Ask students to describe and draw diagrams of each rock (or include photos) in their **Notebook**.

Step 3: Direct students to the online collection *Rocks and Minerals* to look at a wider selection of rocks from the three groups.

Discussion

1. Explain how the features of rocks relate to the way they are formed?



2.1 Rock or mineral

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Minerals are the pure substances, or chemicals, that rocks are made from. Most rocks contain a mixture of several minerals, as individual crystals and grains.
- AC SIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.
- AC SIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- explain the difference between rocks and minerals.

Key vocabulary

Granite, sandstone, calcite, marble, quartz, biotite, plagioclase feldspar, potassium feldspar.

Equipment list

Each STUDENT will require:

- Notebook**

Each GROUP will require:

- samples of granite, sandstone, and calcite
- dissecting microscope

Things to consider

One way of addressing these ideas is to think about minerals as ingredients in a rock recipe. Any given rock is made up of particular constituent minerals, and we can identify the rock on the basis of these minerals. For example, granite has approximately equal proportions of quartz, plagioclase feldspar and potassium feldspar (also called orthoclase feldspar), with other, less abundant accessory minerals. One analogy is that by examining a fruit cake we can see that it contains fruit, and we can evaluate what other ingredients might make up the cake surrounding the fruit. Identifying the fruit and pale-coloured cake illustrates, for example, that it is not a chocolate cake.

The reverse is also true, if you only have the capacity to crystallize (igneous/metamorphic) or accumulate (sediment) certain minerals, then you can only make specific rocks. For example, if you metamorphose a limestone (wholly composed of calcite) then the only rock type you can produce is marble. The analogy is that to make a fruit cake you must use eggs, butter, flour, baking powder and fruit. If you use different ingredients you won't get a fruit cake.



A safety alert was issued regarding the potential for school rock collections to have samples of asbestos. Ensure that you have checked that your collections do not contain any specimens containing asbestos or any ores that may be radioactive.

Teacher content information

The most common rock forming minerals are:

quartz,
calcite,
muscovite,
biotite,
chlorite,
plagioclase feldspar,
potassium feldspar,
augite,
hornblende,
olivine,
kaolinite,
hematite
goethite

Most rocks have minor amounts of other accessory minerals, and some minerals are important to learn about because they are valuable. However, these 13 common rock-forming minerals are the ones that allow us to distinguish one rock from another. Anyone can learn the properties of these minerals, and you should too. These minerals make up almost all igneous rocks and form the main components of metamorphic rocks, or rock weathering products, at the Earth surface. It follows that if we weather and erode these rock types the same constituents will end up in sedimentary rocks. Realising that you can identify most rocks on Earth by learning to identify these 13 minerals is quite profound – so get to it! Remember that the online collection *Rocks and Minerals* is there to support you to do this.

For this activity students should identify that: the granite is a rock composed of quartz, plagioclase feldspar and potassium feldspar with some biotite and muscovite, and possibly some hornblende; the sandstone is a rock likely to contain quartz, some other mineral grains and rock fragments; and the calcite is a mineral (a pure substance) – not a rock.

Key Idea

Minerals are the pure substances, or chemicals, that rocks are made from. Most rocks contain a mixture of several minerals, as individual crystals and grains.

Lesson plan

Step 1: Organise students to work in groups; observing and comparing the three samples.

Step 2: Direct students to the digital link to explore the makeup of rocks and minerals.

Step 3: After students have completed the **Notebook** tasks conclude with a class discussion of students' answers and ensure students have understood the key idea (as above).



2.2 Diagnostic properties

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Minerals are the pure substances, or chemicals, that rocks are made from. Most rocks contain a mixture of several minerals, as individual crystals and grains.
 - All rocks on Earth are made from a limited number of different minerals. Minerals can be identified using a number of observable diagnostic properties.
 - Diagnostic properties are used to distinguish between, and identify, different minerals.
- ACSIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.
- ACSIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- differentiate between objects on the basis of their diagnostic properties, using examples from everyday life
- explain some diagnostic properties that can be used to identify different minerals
- apply diagnostic properties to identify a limited range of minerals.

Key vocabulary

Diagnostic property.

Equipment list

Each STUDENT will require:

- Notebook**

Each GROUP will require:

- mineral samples (e.g. quartz, orthoclase feldspar, calcite, biotite, hornblende, muscovite)
- dissecting microscope
- internet access

Things to consider

Students can use either real mineral samples or selected images from the online collection Rocks and Minerals. The ideal approach would be to use both.

A safety alert was issued regarding the potential for school rock collections to have samples of asbestos. Ensure that you have checked that your collections do not contain any specimens containing asbestos or any ores that may be radioactive.



Teacher content information

Refer to notes from Activity 2.1 and the Rocks and Minerals online collection.

Key Idea

All rocks on Earth are made from a limited number of different minerals. Minerals can be identified using number of observable diagnostic properties.

Lesson plan

Step 1: Look at the peppers and range of cereals and ask students to describe how they differ. Explain that they are employing diagnostic properties to discern differences.

Step 2: Ask students to identify diagnostic properties that could tell apart the six mineral samples. Incorporate these diagnostic properties in their table describing the six minerals.

Step 3: Conclude with a class discussion of the question:

How would you explain the term "diagnostic property" to one of your friends?



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△ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

△ Always ensure safe and ethical disposal of chemicals.

△ Use teacher judgement when considering equipment substitutions.

△ Make considered choices selecting and using resources to minimise waste.

2.3 Testing minerals

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Minerals are the pure substances, or chemicals, that rocks are made from. Most rocks contain a mixture of several minerals, as individual crystals and grains.
 - All rocks on Earth are made from a limited number of different minerals. Minerals can be identified using a number of observable diagnostic properties.
 - Diagnostic properties are used to distinguish between, and identify, different minerals.
- ACSIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.
- ACSIS234 Use scientific knowledge and findings from investigations to evaluate claims based on evidence.
- ACSIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- recognise that the characterisation of minerals is a process that allows differentiation between minerals on the basis of their diagnostic properties
- explain a number of mineral diagnostic properties such as: hardness, cleavage, streak, colour and lustre.

Key vocabulary

Minerals, diagnostic properties, hardness, streak, colour, lustre, cleavage, fracture, acid-fizz, magnetism, fluorescence, specific gravity.

What ideas might your students already have?

Students will use the diagnostic properties of items observed in everyday life (e.g. characteristics of products on a supermarket shelf) to distinguish them from one another for a particular purpose (e.g. finding your preferred brand of cereal), or to distinguish one person from another, but they may not have thought about this skill overtly.

Students may know a little about crystals, minerals and rocks but may not recognise that crystals are minerals, and that rocks are made up of crystals. Students commonly do not understand the difference between minerals and rocks.

Students may not know that they already have many of the skills required for determining the diagnostic properties of minerals.

Equipment list

Each STUDENT will require:

- Notebook**



- **Activity sheet 2.3 Testing minerals** (in *Teacher Guide*)

Each GROUP will require:

- biotite, quartz
- two unknown minerals (suggested – calcite, potassium feldspar)
- mineral study kit
- streak plate
- glass slide, steel paper clip

Things to consider

A safety alert was issued regarding the potential for school rock collections to have samples of asbestos. Ensure that you have checked that your collections do not contain any specimens containing asbestos or any ores that may be radioactive.

Teacher content information

This is a fairly complete list of the defining characteristics of all minerals, provided for teacher background information. Not every test will be used in this activity. The most important idea for students to learn is that every mineral has specific and precise characteristics, just as every pure substance (e.g. water, oxygen, carbon dioxide, lead) has.

The idea is that students are constantly using diagnostic properties to discriminate between people and items, but do not always do so consciously. It is this generic skill that they already have that they will be using to discriminate between different minerals in **Activity 2.3**. Not all of these are included in the activity. Some are not appropriate for safety reasons, such as the taste test, and some are too complex, such as crystal morphology. A complete set is provided here for your background information.

One of the first things you must be very clear about is that minerals are NOT rocks. They are the crystalline materials (crystals) that, when put together in a range of combinations, make up rocks.

That is, they are the component parts that rocks are made of. For example, a basalt rock is formed from the minerals olivine, pyroxene, plagioclase feldspar and a little volcanic glass (a super-cooled liquid like window glass – not a mineral). Another example is granite which is formed from the minerals quartz, potassium-feldspar, plagioclase feldspar, muscovite, some biotite and minor accessory minerals like magnetite (little black flecks).

Brief notes are provided here on mineral diagnostic properties:

Crystal habit: this describes forms that minerals commonly take that may, in part, relate to their crystal morphology (e.g. if they are elongate, blocky, platy). Examples include: the mica family (muscovite (clear), biotite (brown/black), chlorite (green)) that have a platy fabric (like pages in a book) because they have a perfect basal cleavage (see cleavage below); hornblende and tourmaline (elongate crystals); kyanite (blade-shaped crystals – flat and elongate) and more.



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Crystal morphology: the natural crystal shape is commonly well-preserved. This relates to a number of minerals but the most common ones are: pyrite (cube or pentagonal dodecahedron (12-sided ball shape like a soccer ball with each side a pentagon) sometimes called a pyritohedron); garnet (rhombic do-decahedron (12-sided ball shape like a soccer ball with each side a rhombus)); calcite (rhombic prism); halite (rock salt – cubes or rectangular prisms); quartz (hexagonal prism with a pyramid termination); and more.

Hardness: this is considered relative to the Moh's hardness scale that describes a relative scale of hardness using 10 key minerals (*Student Guide* page 15). This is diagnostic for minerals that are very hard (e.g. diamond, corundum) or very soft (e.g. kaolinite, talc, gypsum, calcite), and is commonly used as a comparative measure in Mineral Keys.

Smell: this only relates to a few minerals but the most common ones are sulfur (smells sulfurous) and the ore (economic) mineral arsenopyrite (when struck by a hard object gives off a smell like garlic). Most of the rest smell like soil or sweaty hands.

Taste: this only relates to a few minerals and we would generally NOT encourage students to taste minerals, but if you have some halite (rock salt NaCl) or rarely sylvite (KCl – extremely salty) they might wet their finger, touch the mineral and taste the salt on their finger. The specimen will need to be rinsed under water between students.

Conductivity: this is whether the mineral conducts electricity or not. Set up a simple circuit with a cell (battery), connected by wire to a bulb in turn connected to a wire that can be placed on the mineral; from the other terminal of the cell run another wire that can be placed elsewhere on the mineral. If the mineral conducts the bulb will glow. You can test whether your circuit is working by touching the wires together – the bulb should glow. Sometimes students get false positives for this test because they accidentally touch the wires together instead of testing whether the current flows through the mineral. It is usually the metallic (economic) minerals that conduct electricity (e.g. native copper, galena (lead ore), magnetite and pyrite (iron ore)) and the non-metallic mineral graphite.

Fluorescence: this only relates to a few common minerals, the most common is fluorite. A UV lamp can be set up, preferably in a darkened box, and the fluorite will glow under the UV light.

Streak: this refers to the colour of the mineral powder which may differ from the colour of the mineral. We typically sample this using a 'streak plate', but the non-glazed back of a ceramic tile will serve the same purpose. You draw the mineral across the surface and record the colour of the streak (mineral powder). This is diagnostic for a number of minerals including: pyrite (fool's gold) which has a black streak even though the mineral is a gold colour; hematite (red ochre) which has a red streak even though the rock can look red or silvery grey; and more.

Colour: this is the natural colour of the mineral specimen, for example: malachite (green), azurite (blue), sulfur (yellow), garnet (red or green (grossular garnet)), apatite (pistachio green), olivine (green), kyanite (blue), chlorite (green), biotite (brown/black), muscovite (clear), and more.

Specific gravity: this is how heavy the mineral is relatively speaking and is typically characterized as low, medium or high. Ore minerals for heavy metals like lead (mineral ore is galena) typically have high specific gravity. Delicate, platy minerals like muscovite typically have low specific gravity.

Lustre: this is how earthy (dull), shiny (adamantine), metallic, or translucent/glassy (vitreous) the surface of the minerals are. Lustre is diagnostic for a number of minerals including: pyrite, galena, chalcopyrite (all metallic); quartz (vitreous), kaolinite (earthy/dull), diamond (adamantine), and more.



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Optical properties: this is diagnostic for a few minerals, the most common is calcite. If you draw a dot on a piece of paper and place a translucent rhombic prism of calcite over the top and look through it you will see two dots because the calcite crystal refracts the light (like a prism).

Cleavage: this describes the fractures (aligned with the lattice arrangement of elements that make up the mineral) along which certain minerals will naturally break.

Fracture: how readily the mineral will fracture and what shapes it will fracture into: conchoidal fracture – a curved fracture as in broken glass (e.g. obsidian glass, opal, quartz); hackly fracture – an irregular or jagged fracture not along any crystal planes (e.g. in native metals like copper, silver); earthy fracture – looks like freshly broken soil (e.g. ilmenite, kaolinite, some goethite (yellow ochre)); splintery fracture – fractures into elongate shapes (e.g. kyanite, chrysotile – a form of asbestos).

Magnetism: whether or not the mineral is magnetic, and therefore whether it can attract metallic materials to it (e.g. the mineral magnetite can pick up nails etc.). One test is to see if the material can move the needle of a compass – this will detect even low levels of magnetism observed in many metals.

Lesson plan

Step 1: Using the *Student Guide* introduce students to each of the diagnostic tests for minerals.

Step 2: In small groups have students conduct diagnostic tests on quartz and biotite and record their results.

Step 3: Compare their results against the table in **Activity sheet 2.3 Testing minerals**.

Step 4: Ask students to test each of the two unknown minerals (suggested – calcite, potassium feldspar) and complete the rows in the bottom of the table in **Activity sheet 2.3 Testing minerals**.

Step 5: Explore the other minerals in the mineral study kit. Test them and compare results with the table in **Activity sheet 2.3 Testing minerals**.

Discussion:

1. Explain the properties you used to 'fingerprint' the 'unknown' minerals.



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2.4 Rock identification

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Minerals are the pure substances, or chemicals, that rocks are made from. Most rocks contain a mixture of several minerals, as individual crystals and grains.
 - All rocks on Earth are made from a limited number of different minerals. Minerals can be identified using a number of observable diagnostic properties.
 - Diagnostic properties are used to distinguish between, and identify, different minerals.
- ACSIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.
- ACSIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- recognise that a rock can be identified by characterising the minerals it contains
- apply their understanding of mineral characteristics to identify some key rock types.

Equipment list

Each STUDENT will require:

- Notebook**
- Internet access
- Activity sheet 2.3 Testing minerals** (in *Teacher Guide*)

Teacher content information

Specific rock types can be identified by the minerals they are comprised of. However, successful identification of the minerals present in many rocks can be challenging. For this reason this activity provides a heavily scaffolded approach through a student digital activity.

Key Idea:

Specific rock types are always comprised of the same rock forming minerals. Identification of these minerals can be used to identify the rock.

ROCK	MINERALS PRESENT	NAME OF ROCK
A	Quartz	Sandstone
B	Potassium feldspar, plagioclase feldspar, quartz, biotite	Granodiorite
C	Calcite	Limestone



Lesson plan

- Step 1:** Direct students from the *Student Guide* to the digital link to view Rocks A, B and C. Use **Activity sheet 2.3 Testing minerals** to identify each of the minerals from their diagnostic properties.
- Step 2:** Use the table shown in the **Notebook** activity to identify each of the rocks from their mineral makeup.
- Step 3:** Discuss students answers to the **Notebook** tasks and emphasise the key idea that:
‘Specific rock types are always comprised of the same rock forming minerals.
Identification of these minerals can be used to identify the rock.’



2.5 Mining our everyday minerals

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Minerals are the pure substances, or chemicals, that rocks are made from. Most rocks contain a mixture of several minerals, as individual crystals and grains.
 - All rocks on Earth are made from a limited number of different minerals. Minerals can be identified using a number of observable diagnostic properties.
- ACSHE 134 Scientific knowledge has changed peoples' understanding of the world and is refined as new evidence becomes available.
- ACSHE 135 Solutions to contemporary issues that are found using science and technology, may impact on other areas of society and may involve ethical considerations.
- ACSHE136 People use science understanding and skills in their occupations and these have influenced the development of practices in areas of human activity.
- ACSI148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- describe a range of products that are made from minerals
- match some minerals with their uses
- appreciate the value of minerals and fossil fuels to the Australian economy.

Key vocabulary

Ore, bauxite, aluminium, titanium, hematite, iron, steel, quartz, glass, bricks, clay, concrete, fossil fuels, petrochemicals.

Equipment list

Each **STUDENT** will require:

- Notebook**
- Student Guide*
- Internet access

Things to consider

As stated at **Activity 1.1**, if we understand why we might want to identify rocks and minerals in different places, it makes learning their properties more relevant to the students. This also allows us to address the **Science as a Human Endeavour** aims for this subject area.

One of the main reasons we want the students to learn about minerals is because we use them in our daily lives. They are typically the raw materials that many of our material items (e.g. cars, telephones) are made of, or the energy supplies (e.g. coal, oil) that we use to make them operate.



This lesson introduces students to the concept that materials like glass and steel are made from the minerals we have extracted from rocks. Students are commonly fascinated by the wide range of applications we have for rocks and minerals. For example kaolinite clay is used in the pottery and brick-and tile-making industries. It is also used as a thickener in paints, as a filler in high quality paper, as a base for mineral cosmetics, and to add non-toxic bulk to thick shakes in the food industry.

Teacher content information

Refer to the online collection *Rocks and Minerals* that illustrates common uses for each rock and mineral detailed.

Lesson plan

- Step 1:** Show students three objects (perhaps cups) made from glass, aluminium and iron. Ask them: Where did we get these materials from and how?
- Step 2:** Present the *Student Guide* and show the video which explains the production of iron, aluminium and Portland cement (limestone and clay).
- Step 3:** Ask students to test their knowledge in the click-and-drag activity linking products with the mineral they are made from.
- Step 4:** Direct students to the **Notebook** activity. Here students interpret a pie chart to answer questions about Australia's export industry.
- Step 5:** You may like to set as homework the interactive in the **Find out more** section which describes the minerals in common household objects.



2.6 Careers in mineral exploration

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Minerals are the pure substances, or chemicals, that rocks are made from. Most rocks contain a mixture of several minerals, as individual crystals and grains.
- ACSHE 134 Scientific knowledge has changed peoples' understanding of the world and is refined as new evidence becomes available.
- ACSHE 135 Solutions to contemporary issues that are found using science and technology, may impact on other areas of society and may involve ethical considerations.
- ACSHE136 People use science understanding and skills in their occupations and these have influenced the development of practices in areas of human activity.
- ACSI148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- describe some of the processes and steps involved in the discovery and mining of valuable mineral deposits
- describe some careers involved in the discovery and mining of minerals.

Key vocabulary

Seismic, geologist, geochemist, geophysicist.

Equipment list

Each STUDENT will require:

- Notebook**
- Student Guide*
- Internet access

Things to consider

In Australia our highest income earner is the export of minerals including commodities like coal and iron ore, and metals like gold. Students may not have realised that Australia is so reliant on these exports. This might be an interesting opening to the following lesson.

Give some thought to how you want students to view the videos and explore the rollovers about different careers. You might like to watch some of the videos on geologist careers as a class before letting students work individually or in pairs to explore the sets of the resources.



Teacher content information

Because we need to find specific minerals for particular applications, we need to use our understanding of the processes that form rocks to help us focus our search for these 'economic' minerals.

For example, if we want to find a source for salt, we would look in a desert area that has salt lakes or had them in the geological past, or look in a littoral swamp area near the coast where sea water was currently evaporating, or was in the geological past. We are not likely to look in an area of rainforest for a salt deposit, because the high rainfall means that this is a poor environment for salt deposition. Have students think about the kinds of skills required to understand these processes, and then identify where we might look for mineral resources.

Once we have thought about what we are looking for, we need to identify where to look, how we might find these resources, and how we might extract them from the ground.

If the minerals we are looking for are just one component in a rock, we have to think about how we are going to separate the part that we want, from the rest of the minerals making up the rock. Does this separation happen naturally in nature (as in the salt example), or do we have to use a chemical process to separate the useful material from the balance of the rock?

Finally, there might be multiple uses for a particular mineral. What influences which process they might be used for? This includes the quality of the resource, whether it is pure enough for a particular industrial process, and how close it is to where it is needed (e.g. the export of iron ore to China as a raw resource).

Lesson plan

- Step 1:** You might like to begin this lesson with a brainstorm question: 'What kind of jobs do people do in the mining sector?'
- Step 2:** View the videos about geologist careers as a class or individually or in pairs.
- Step 3:** Direct students to explore the rollover comments and popup boxes of geoscientists.
- Step 4:** Read the section about Mineral exploration before opening the interactive. It describes the order of tasks in mineral exploration.
- Step 5:** Open the interactive and first explore what useful geological deposits are hidden under the landscape. Then work out the order of events in a mineral exploration process.
- Step 6:** Complete the **Notebook** tasks. Discuss completed answers with the class.
- Step 7:** You might like your students to explore where Australia's mines are located in the **Find out more** section.



2.7 Money for rocks

Curriculum links

- ACSHE136 People use science understanding and skills in their occupations and these have influenced the development of practices in areas of human activity.
- ACSIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- identify some minerals that are most important for Australian exports.

Key vocabulary

Diamonds, iron ore, banded iron formation, chalcopryrite, copper, coal.

What ideas might your students already have?

- Students might identify gold and diamonds as having more value because they are expensive items.
- Students might recall from **Activity 2.5** that iron ore is a major Australian export.

Equipment list

Each **STUDENT** will require:

- **Notebook**
- *Student Guide*
- Internet access

Things to consider

This activity could be set as homework.

Teacher content information

The key idea here is that minerals that seem to be most valuable, such as gold and diamonds, do not actually account for the majority of Australia's export income.

Lesson plan

Step 1: It may be useful to include a brief class discussion about which minerals appear to be most valuable to Australia; diamonds, gold, iron, copper or coal.

Step 2: Allow students to explore the rollover comments and discover which mineral is worth the most income to Australia.



3.1 Big and small crystals

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Plutonic igneous rocks form when magma is intruded below the Earth surface. The magma cools slowly because it is insulated by the enclosing rock, and there is time for large crystals to grow forming a coarse-grained rock.
 - Volcanic igneous rocks form when magma is erupted at the Earth surface. The magma cools rapidly because there is a large temperature contrast between the magma and the air, so the rock is quenched forming a fine-grained rock.
- ACSIS140 Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed.
- ACSIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.
- ACSIS234 Use scientific knowledge and findings from investigations to evaluate claims based on evidence.
- ACSIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- identify that crystal growth is affected by temperature.

Key vocabulary

Crystalline, glass, quartz, phenyl salicylate.

Equipment list

Each STUDENT will require:

- **Notebook**
- *Student Guide*

Each GROUP will require:

- fine crystals of phenyl salicylate
- sample of molten salol (in hot water bath)
- 2 glass slides
- dissecting or compound microscope with low power
- pipette, paper towel



Things to consider

Phenyl salicylate (Salol) $C_{13}H_{10}O_3$ melts at $42^{\circ}C$. Avoid contact with skin and eyes. Ensure students wear protective glasses and disposable gloves when handling this chemical.

This activity needs to be carried out quite quickly so that the slides do not warm up or cool down too much before students make their observations. It may be necessary for students to practice the experimental protocol a few times first.

Make sure that you have tried this activity yourself, using the materials the students will be using, before completing this activity in class. This will allow you to troubleshoot any experimental issues, relating to slide temperature and the type of salol you have, more effectively. There are several different versions of this little experiment around, but we have tested these and found that this protocol provides the most interesting visual experience for the students.

A safety alert was issued regarding the potential for school rock collections to have samples of asbestos. Ensure that you have checked that your collections do not contain any specimens containing asbestos or any ores that may be radioactive.

Note: Do not dispose of solutions of the binders down the sink as they may set in the pipes and cause a major plumbing problem.

Teacher content information

The concept here is that if magma (molten rock) cools slowly (e.g. if it is intruded below the Earth surface) there is sufficient time for large crystals to grow. The result is a coarse-grained igneous rock like granite or gabbro.

If magma is cooled rapidly (e.g. when magma is erupted at the Earth surface) it is quenched rapidly and there is limited time for crystals to grow. The result is a fine-grained igneous rock like rhyolite or basalt.

Key Idea:

Plutonic igneous rocks form when magma is intruded below the Earth surface. The magma cools slowly because it is insulated by the enclosing rock, and there is time for large crystals to grow forming a coarse-grained rock. **Volcanic** igneous rocks form when lava is erupted at the Earth surface. The lava cools rapidly because there is a large temperature contrast between the lava and the air, so the rock is quenched forming a fine-grained rock.

Lesson plan

Step 1: Inform students of the safety considerations of this experiment.

Step 2: You might consider demonstrating to students how to prepare the two slides. Have students follow the instructions outlined in the *Student Guide*.

Step 3: Allow students time to watch the crystals grow.

Step 4: Students should record their observations in their **Notebook**. Direct them to the discussion questions. Discuss with the class their answers.

Step 5: Look at the chemical structures of quartz and glass shown on page 26 in the *Student Guide*. Ask students to note the difference between these structures.



3.2 Erupting and intruding

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Plutonic igneous rocks form when magma is intruded below the Earth surface. The magma cools slowly because it is insulated by the enclosing rock, and there is time for large crystals to grow forming a coarse-grained rock.
 - Volcanic igneous rocks form when magma is erupted at the Earth surface. The magma cools rapidly because there is a large temperature contrast between the magma and the air, so the rock is quenched forming a fine-grained rock.
- ACSIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- identify two different types of igneous rock (basalt and granite)
- explain how igneous rocks are formed
- explain the link between crystal size and formation of igneous rocks.

Key vocabulary

Magma, plutonic, intrusive, crystals, granite, lava, volcanic, extrusive, basalt.

Equipment list

Each **STUDENT** will require:

- **Notebook**
- *Student Guide*
- Internet access

Each **GROUP** will require:

- samples of granite and basalt
- hand lens and/or dissecting microscope

Teacher content information

Refer to the notes at **Activity 3.1**.



Lesson plan

- Step 1:** Take students through the information provided in the *Student Guide*.
- Step 2:** Provide students with samples of granite and basalt to observe using the hand lens and/or dissecting microscope. Ask students to write descriptions in their **Notebooks**.
- Step 3:** Explore the formation of igneous rocks by watching the videos. Compare volcanic rocks with plutonic rocks in the online collection *Rocks and Minerals*.
- Step 4:** Ask students to complete the on-line quiz to test their understanding of plutonic and volcanic rocks.
- Step 5:** You might consider setting for homework the task of exploring the basalt formation Fingals Cave in the *Find out more* section.



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⚠ Always ensure safe and ethical disposal of chemicals.

⚠ Use teacher judgement when considering equipment substitutions.

⚠ Make considered choices selecting and using resources to minimise waste.

4.1 Breaking rocks

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Rocks can be broken down through a combination of chemical and physical processes. These processes are referred to as weathering.
 - Sediments created through the processes of weathering can be moved from one place to another by a variety of physical agents, including ice, water and wind. This process is called erosion.
 - Different parts of the landscape are characterized by different sediments because the processes taking place there contribute to specific sediment types.
- AC SIS139 Identify questions and problems that can be investigated scientifically and make predictions based on scientific knowledge.
- AC SIS140 Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed.
- AC SIS141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.
- AC SIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.
- AC SIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.
- AC SIS146 Reflect on scientific investigations including evaluating the quality of the data collected, and identifying improvements.
- AC SIS234 Use scientific knowledge and findings from investigations to evaluate claims based on evidence.
- AC SIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- explain how rocks can be broken down through the processes of physical and chemical weathering
- explain the effects of surface area and temperature on chemical weathering
- distinguish between the processes of weathering and erosion.

Key vocabulary

Physical weathering, chemical weathering, surface area, erosion, sediments, onion skin weathering, tors.

Equipment list

Each STUDENT will require:

- **Notebook**



Each GROUP will require:

- 9 sugar cubes
- medium glass jar for shaking the cubes
- 2 beakers
- warm and cold water
- access to electronic balance

Things to consider

Students will not be used to thinking about rocks as objects that can be broken down. It would be useful to initiate a class discussion about where rocks might come from and what happens to them over time to elicit their understanding of these topics.

It is important for students to understand the idea that weathering is the chemical and physical breakdown of the rock, and that erosion is the transport of the weathered material away from the place it was formed. This is illustrated in the video linked to this activity. Take care not to confuse the words abrasion (physical weathering of rock) with erosion (transport of weathered materials).

Some students may draw the link between the second part of the practical and prior learning about surface area that they have been taught in chemistry. This is an opportunity to ensure that they are not compartmentalising between subject areas and that they appreciate that the learning in different aspects of science is complementary.

Teacher content information

Physical weathering is the abrasion of a rock. Chemical weathering is the dissolution of a rock. These processes commonly happen in parallel. Media that facilitate the weathering/breakdown of rocks, including wind, water and ice, are known as weathering agents.

Key Ideas:

Rocks can be broken down through a combination of chemical and physical processes. These processes are referred to as weathering. Sediments created through the processes of weathering can be moved from one place to another by a variety of physical agents, including water, glaciers and wind. This process is called erosion.

Lesson plan

Part 1: Physical weathering

Form students into groups and conduct the experiment as described in the student guide. After each shaking trial remove the five largest remaining cube pieces for photographing or drawing and then weighing.

Part 2: Chemical weathering

Allow time for students to discuss in their group how they will design this experiment. Anticipate that students might ask for equipment to crush the cubes, stopwatches and possibly thermometers. As a class discuss the conclusions from both experiments.

Digital activities

Direct students to exploring the rollovers on each of the images.

You might like to project the video onto a screen for the class as a whole and then ask students to complete the **Notebook** questions. Consider demonstrating the weathering ability of ice by performing the experiment shown in the *Find out more* section. Students will be eager in the following lesson to see the results of the experiment.



4.2 Which sediments where?

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Sediments created through the processes of weathering can be moved from one place to another by a variety of physical agents, including ice, water and wind. This process is called erosion.
 - Different parts of the landscape are characterized by different sediments because the processes taking place there contribute to specific sediment types.
 - Clastic sedimentary rocks are formed from grains that are cemented together. The characteristics of the grains in each rock relate to the conditions under which the sediments were first formed, and what they have experienced since.
- ACSI145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.
- ACSI148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- relate the formation of different sediments to different landscapes
- relate different sediments to the formation of different sedimentary rocks.

Key vocabulary

Sediment, mudstone, sandstone, conglomerate, limestone, halite, glacier, estuary, marine, ferricrete, calcrete, silcrete.

What ideas might your students already have?

Students will be familiar with sediments in particular environments (e.g. sand at the beach, pebbles in rivers) but may not recognise that they are generated and/or deposited in a much wider range of environments.

Teacher content information

Different parts of the landscape are characterised by different sediments because the processes taking place there contribute specific sediment types. These are referred to as sedimentary environments. High parts of mountain ranges typically have poorly-sorted, angular, coarse sediments that have formed locally (immature sediment). River mouths typically have well-sorted, rounded, fine sediments that have been transported a long distance from the original site of formation (mature sediment).

Key Idea:

Different parts of the landscape are characterised by different sediments because the processes taking place there contribute specific sediment types.



Lesson plan

- Step 1:** Show students the various sedimentary environments in the *Student Guide*. What forms of weathering are taking place in each? What physical agents (water, glacier, and wind) are moving the sediments?
- Step 2:** Direct students to the digital activity to explore the rollovers of each sedimentary environment and then to the **Notebook** activity.
- Step 3:** Discuss the local places students have identified as being likely locations to find various sediments that could ultimately form into sedimentary rocks.



⚠ As a teacher, it is your responsibility to follow your school safety and risk, policies and procedures.

⚠ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

⚠ Always ensure safe and ethical disposal of chemicals.

⚠ Use teacher judgement when considering equipment substitutions.

⚠ Make considered choices selecting and using resources to minimise waste.

4.3 Sticking things together

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Clastic sedimentary rocks are formed from grains that are cemented together. The characteristics of the grains in each rock relate to the conditions under which the sediments were first formed, and what they have experienced since.
- AC SIS139 Identify questions and problems that can be investigated scientifically and make predictions based on scientific knowledge.
- AC SIS140 Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed.
- AC SIS141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.
- AC SIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.
- AC SIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.
- AC SIS146 Reflect on scientific investigations including evaluating the quality of the data collected, and identifying improvements.
- AC SIS234 Use scientific knowledge and findings from investigations to evaluate claims based on evidence.
- AC SIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- explain how sediments can form into rocks in the presence of suitable binders
- design a method for testing and comparing the hardness of hand-made rocks.

Equipment list

Each **GROUP** will require:

- selection of gravels, sands and silt
- selection of possible binders including:
 - PVA glue
 - gypsum
 - cement powder
 - plaster of Paris
 - Calcium hydroxide (slaked lime)
 - Calcium carbonate
- small beaker (200 mL)
- plastic cups with holes in the bottom, and spoons



Each STUDENT will require:

- **Notebook**
- camera (if available)

Things to consider:

Students will find this a very engaging activity and will be tempted to begin mixing materials haphazardly in a process of exploration. Take the time at the beginning of the lesson to draw students' thinking to the following:

- Task 1 - to determine which binder material produces the most durable rock.
- Task 2 - to design a test to compare the hardness of each of the rocks.
- Identify which variables must be controlled.
- How can you compare rock hardness that will give you quantitative measurements?
- How can the results be presented in a table and also displayed in a graph?

Allow students time to discuss in groups the answers to these questions before beginning the experiment.

In designing their test for rock hardness students could be encouraged to think of what sorts of conditions most rocks have to endure. These might include physical pressure, due to other rocks being piled on top, and resistance to water.

Teacher content information:

Sediments lithify (solidify) into sedimentary rock due to a process called diagenesis, that typically involves compaction and cementation of sediment particles (clasts) together. This generates sedimentary rocks known as clastic sedimentary rocks.

The more compacted and the better cemented (dependent on cement type) the sedimentary rock, the more resistant this rock is to subsequent weathering and erosion.

Lesson plan

Part 1

Step 1: Using the *Student Guide*, introduce this guided inquiry activity and show students the different materials they will be handling.

Step 2: Working in student groups draw students' thinking to the following:

- Task 1 - to determine which binder material produces the most durable rock.
- Task 2 - to design a test to compare the hardness of each of the rocks.
- Identify which variables must be controlled.
- How can you compare rock hardness that will give you quantitative measurements?
- How can the results be presented in a table and also displayed in a graph?

Step 3: Make a range of 'rocks'. You may leave the cups overnight to allow sufficient time for the rocks to 'set'.

Part 2

Step 1: Once the rocks have set students can test each of their rocks for hardness. Take photographs of each rock.

Step 2: Ask students to record quantitative measurements that can be presented in both table and graph formats in their **Notebooks**.

Step 3: Add photographs to illustrate their reports and ask students to draw conclusions from their tests.



4.4 Grains and rocks

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Clastic sedimentary rocks are formed from grains that are cemented together. The characteristics of the grains in each rock relate to the conditions under which the sediments were first formed, and what they have experienced since.
- ACSIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.
- ACSIS146 Reflect on scientific investigations including evaluating the quality of the data collected, and identifying improvements.
- ACSIS234 Use scientific knowledge and findings from investigations to evaluate claims based on evidence.
- ACSIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- recognise that different sedimentary rocks can be characterised by the size of their grains
- recognise that different sedimentary rocks can be characterised by whether they form from grains or chemical or living sediments.

Key vocabulary:

Clast, conglomerate, grain size comparator

Equipment list

Each GROUP will require:

- specimens of sandstone, mudstone and conglomerate
- specimens of beach sand, dried mud or silt, pebbles
- dissecting microscope and hand lens
- scissors

Each STUDENT will require:

- internet access
- **Activity sheet 4.4 Grain size comparator**
- **Notebook**



Teacher content information:

The grains in rocks are called clasts. Any sedimentary rock that is comprised of grains cemented together is called a 'clastic' rock. The grains in different rocks are characterised by the nature of the sediments from which the rocks are formed. These, in turn, are characterised by how the sediments were formed in the first place, and what they have experienced since.

Lesson plan

- Step 1:** Students may cut out their Grain size comparators from **Activity sheet 4.4**. Focus their attention to the sizes of clasts and the names: silt, sand, granules, pebbles, cobbles, and boulders. The abbreviation μm is for micron or micrometre. There are 1 million (1,000,000) μm in a metre and there are 1000 μm in 1 mm.
- Step 2:** Using the *Student Guide*, take the class through the steps involved in this activity. Students may copy the results tables into their **Notebook** or use the **e-Notebook**.
- Step 3:** Students may choose to characterise the 3D specimen of conglomerate. It is comprised of clasts from fine sand through to large pebbles.
- Step 4:** Ask students to determine which sedimentary environment produced each rock. Students might benefit for revisiting **Activity 4.2 Which sediments where?**
- Step 5:** Show the class the videos on sedimentary rock formation. After watching Sammy Sandstone ask students to complete the **Notebook** tasks.
- Step 6:** Invite students to explore the remaining digital activities. The *Find out more* section has a link to the online collection *Rocks and Minerals* where students can see the sedimentary rocks are classified as either clastic, biogenic or chemical.



4.5 Making rocks from body parts?

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Biogenic rocks are formed from sediments of accumulated body parts. Key examples are coal and limestone.
- ACSHE 135 Solutions to contemporary issues that are found using science and technology, may impact on other areas of society and may involve ethical considerations.
- ACSIS141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.
- ACSIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- explain how living material can be incorporated into sediments and transformed into sedimentary rocks
- use the acid fizz test to identify limestone and related rocks
- describe some current uses of rocks made from living material.

Key vocabulary:

Limestone, coal, calcium carbonate (CaCO_3), hydrochloric acid (HCl), peat, biogenic rocks.

Equipment list

Each GROUP will require:

- piece of limestone, some egg shell, small amount of calcium carbonate, some sea shells fragments, sample of sand and a piece of sandstone
- dissecting microscope, hand lens
- dropper bottle of dilute HCl
- small piece of coal

Each STUDENT will require:

- Notebook**

Things to consider:

If possible, find some display specimens of coal and limestone that have visible fossils (plant and animal remains respectively) in them. This is useful for students to make the connection that these are different from clastic rocks. Take care though – there are some clastic rocks (e.g. sandstones) that also have fossils in them – but these will not illustrate the biogenic origin of the sediments that we are striving for. Once again it is better to have multiple specimens of samples that look the same (class sets of a few examples) rather than an array of specimens that look different to each other.



Although the acid is dilute, it will sting open skin or eyes.

Teacher content information:

Biogenic sediments are formed from accumulations of body parts (organic material). The two main ones that you want students to recognise are coal formed from plant matter and limestone formed from animal remains.

Key Idea:

Biogenic rocks are formed from sediments of accumulated body parts. Key examples are coal and limestone.

Lesson plan

Part 1

Step 1: Warn students about the dangers of acid. Students should test each sample and note in their Notebooks any observations (sight, sound, smell).

Step 2: Observe the limestone for individual grains using the dissecting microscope or hand lens.

Part 2

TEACHER DEMONSTRATION IN FUME HOOD'

Step 1: Observe the coal sample under the dissecting microscope and describe its appearance.

Step 3: Let students explore the digital activities on limestone caves and fossils before attempting the **Notebook** tasks.



5.1 These rocks are so hot!

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Contact metamorphism occurs when rocks are 'baked' at elevated temperature when hot magma is intruded adjacent to them.
- ACSIS139 Identify questions and problems that can be investigated scientifically and make predictions based on scientific knowledge.
- ACSIS140 Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed.
- ACSIS141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.
- ACSIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.
- ACSIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.
- ACSIS146 Reflect on scientific investigations including evaluating the quality of the data collected, and identifying improvements.
- ACSIS234 Use scientific knowledge and findings from investigations to evaluate claims based on evidence.
- ACSIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- explain the role of heat in the formation of contact metamorphic rocks.

Key vocabulary:

Metamorphic rock, parent rock, contact metamorphism, hornfels, ceramics, kiln.

Equipment list

Each GROUP will require:

- some potters clay
- metal tongs
- access to a clay firing oven, special heat gloves; or
- access to a regular oven, baking tray and oven mitts

Each STUDENT will require:

- **Notebook**



Things to consider:

This activity will need to run over two lessons because the clay bricks need time to completely dry. Ensure clay bricks are completely dry to avoid risk of explosion. Note: consult with the art (pottery specialist) to advise firing temperature, and because ovens may vary, experiment with oven cooking temperatures in advance to ensure heating time is suitable.

Students are familiar with bricks and tiles, or ceramic pottery (bowls, toilets) but may not have realised that these are essentially made from baked clay.

Teacher content information:

Baking clay bricks is a sound analogy for contact metamorphism, where there is elevated temperature at relatively low pressure.

Key Idea:

Contact metamorphism occurs when rocks are 'baked' at an elevated temperature when hot magma is intruded adjacent to them.

Lesson plan

Part 1

Step 1: Decide on your method of firing the clay.

Step 2: Instruct your students to make small bricks for example 1cm by 1cm by 1cm. This way a class set of bricks will fit on the baking tray, storage and handling of the bricks is more manageable.

Step 3: Students can write their initials into the bricks for future identification before leaving them aside in the lab to dry out.

Step 4: Direct students to the section on contact metamorphism in the *Student Guide* and the video on tile making.

Part 2

Step 5: Instruct students on the safety aspects of hot bricks including the use of special heat gloves, the importance of not touching the hot bricks as soon as they are removed from the oven, and handling bricks using metal tongs.

Step 6: While the bricks are cooling students should design their tests to determine how their bricks can withstand water and pressure. Carry out the tests and ensure students record their observations in their **Notebook**.

Step 7: Direct students to the **Notebook** activity. (This could be set as a homework exercise.)



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▲ Always ensure safe and ethical disposal of chemicals.

▲ Use teacher judgement when considering equipment substitutions.

▲ Make considered choices selecting and using resources to minimise waste.

5.2 Regional metamorphism

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Metamorphic rocks can be created through processes involving both high temperature and high pressure. These conditions typically occur deep within the Earth's crust.
- AC SIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.
- AC SIS234 Use scientific knowledge and findings from investigations to evaluate claims based on evidence.
- AC SIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- explain the role of temperature and pressure in the formation of regional metamorphic rocks
- identify that metamorphic rocks can have similar chemical compositions to their parent rock.

Equipment list

Each GROUP will require:

- samples of limestone and marble
- dilute hydrochloric (0.5M) acid in a dropper bottle

Each STUDENT will require:

- Notebook**

Things to consider:

It is best to use a fairly loosely consolidated limestone for this exercise so that it looks quite different structurally than marble. This illustrates the recrystallization process caused by the metamorphic process, because the marble is much denser than the limestone.

A safety alert was issued regarding the potential for school rock collections to have samples of asbestos. Ensure that you have checked that your collections do not contain any specimens containing asbestos or any ores that may be radioactive.

Teacher content information:

Marble is the metamorphosed equivalent of limestone. It is formed when the parent limestone has been subjected to elevated temperature and pressure. Most examples of metamorphic rocks, particularly large beds, have experienced both high temperatures and high pressures. This occurs as a result of being buried deep within the crust.



Lesson plan

- Step 1:** In the *Student Guide* introduce students to the famous tilted beds of metamorphic rocks in northern Scotland that extend for about 250 km.
- Step 2:** Students conduct simple acid fizz tests on limestone and marble to see that they have similar chemical properties. Students complete the **Notebook** activity related to this experiment.
- Step 3:** Direct students to the digital click-and-drag activity to show the correct order of events from a coral reef to Michelangelo's David.
- Step 4:** Allow students to explore the metamorphic rocks held in the online collection *Rocks and Minerals*.



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⚠ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

⚠ Always ensure safe and ethical disposal of chemicals.

⚠ Use teacher judgement when considering equipment substitutions.

⚠ Make considered choices selecting and using resources to minimise waste.

5.3 A tortured history

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Different metamorphic rocks can form under a range of different conditions of heat and pressure.
- ACSHE136 People use science understanding and skills in their occupations and these have influenced the development of practices in areas of human activity.
- ACSI148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- provide a summary of the conditions under which metamorphic rocks form and the processes involved
- explain that if metamorphic rocks melt then they are becoming magma and are no longer metamorphic rocks.

Things to consider

A safety alert was issued regarding the potential for school rock collections to have samples of asbestos. Ensure that you have checked that your collections do not contain any specimens containing asbestos or any ores that may be radioactive.

Teacher content information:

If any rock materials experience elevated temperature (without melting) or pressure, or both, the rocks will change form or metamorphose. The rock types that are generated depend on what was metamorphosed in the first place, and the combination of pressure and temperature they have been subjected to.

These fall into three main categories:

Contact metamorphism (high temperature, low pressure) occurs where rocks are essentially baked because hot rocks have been emplaced next to them.

Regional (burial) metamorphism occurs when both pressure and temperature increase in parallel, typically because the rocks have been progressively buried.

Finally there are some very localised areas, where rocks are subjected to elevated pressure without necessarily experiencing higher temperatures, usually in fault zones. This is called high pressure or cataclastic metamorphism, but we do not usually teach this in great detail because students rarely encounter these rocks in the field.

Metamorphic rocks have a characteristic layering or foliation (platiness). Gneisses have a special segregation of minerals into lighter and darker layers, and this is the main feature that distinguishes them from schists. If they continue to be compressed and heated they will eventually melt, forming an igneous rock. An unusual rock type that preserves partially melted gneiss is called a migmatite. It has the appearance of gneiss blocks enclosed in granite. Once the gneiss is completely melted the new rocks formed will be igneous rocks.



Key Idea:

Different metamorphic rocks can form under a range of different conditions of heat and pressure. Contact metamorphism (increased temperature) changes shale into hornfels. Burial metamorphism (increased temperature and pressure) changes shale progressively into slate, schist and gneiss.

Lesson plan

Step 1: Watch the three short videos.

Step 2: Complete answers to the **Notebook** questions. Discuss the answers students have written.



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△ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

△ Always ensure safe and ethical disposal of chemicals.

△ Use teacher judgement when considering equipment substitutions.

△ Make considered choices selecting and using resources to minimise waste.

6.1 Rocks into rocks

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Although most rocks appear to be endlessly durable they have a life story. Rocks are broken down and rebuilt over very long periods of time.
 - All rocks on Earth can be broadly categorized into three groups; sedimentary, igneous and metamorphic. These groups of rocks have different characteristics that relate to the manner in which they are formed.
 - Rocks can be broken down through a combination of chemical and physical processes. These processes are referred to as weathering.
 - Rock materials are not destroyed but cycle through the processes of formation of sedimentary, igneous and metamorphic rocks, over millions of years.
- ACSHE226 Science knowledge can develop through collaboration across the disciplines of science and the contributions of people from a range of cultures.
- ACSHE136 People use science understanding and skills in their occupations and these have influenced the development of practices in areas of human activity.
- AC SIS146 Reflect on scientific investigations including evaluating the quality of the data collected, and identifying improvements.
- AC SIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- explain the concepts and processes described in each step of a rock cycle diagram.

Key vocabulary:

Rock cycle, processes, compaction, cementation, weathering, erosion, metamorphosis, melting, solidification, sediments, lava, magma.

Equipment list

Each GROUP will require:

- wax crayons in 3 contrasting colours
- sharp knife or grater (or pencil sharpener – see below)
- aluminium foil (or metal cupcake cases)
- very hot water
- rolling pin or heavy book
- candle
- iced water



- kitchen paper
- tongs

Each STUDENT will require:

- **Notebook**
- internet access

Teacher content information:

The notes here are largely in common with **Activity 1.3**.

Essentially the rock cycle reinforces the concept that matter is neither created nor destroyed but is transformed into different configurations. This is very big idea in science. In summary:

If any rock materials are weathered and eroded they will form sediment, which can be cemented and compacted to become a (clastic) sedimentary rock. Other sedimentary rocks form either by precipitation from a supersaturated solution (chemical), or due to the accumulation of plant or animal matter (biogenic).

If any rock materials are subjected to sufficient heat, the composite minerals melt and magma (hot molten rock) forms. When this magma cools it forms igneous rocks. If magma cools slowly below the surface there is time for larger crystals to form and the rocks are called plutonic rocks (e.g. granite). Magma may erupt at the surface to violently eject block and ash material (volcanic sediments) or it may erupt to produce lava flows which cool quickly to form fine-grained volcanic rocks (e.g. basalt).

If any rock materials (sedimentary, igneous, or metamorphic) experience elevated temperature (without melting) or pressure, or both, the rocks will change form or metamorphose. The rock types that are generated depend on what was metamorphosed in the first place, and the combination of pressure and temperature they have been subjected to.

The other concept that students (and most adults) struggle to truly comprehend, is geological time. The span of geological time is immense, however, if we focus on that component of the geological record that is relevant to the processes we are learning about in a particular area, this makes working with geological time a little more manageable.

Key Idea:

Although most rocks appear to be endlessly durable they have a life story. Rocks are broken down and rebuilt over very long periods of time.

Lesson plan

Step 1: Discuss with students the rock cycle diagram in the *Student Guide* which integrates all of the concepts studied in this unit.

Step 2: Invite students to try the *Rock Cycle Diagram* interactive and test their understanding of rock types and geological processes.

Step 3: Direct students to the practical activity where students simulate geological processes using grated crayons. This activity allows students to elaborate on their understanding of the rock cycle.

Step 4: Follow with the **Notebook** tasks which invite students to research the geology of the local area and elaborate about its context in the rock cycle.

Step 5: You might consider setting the *Find out more* task for homework.



6.2 On the level

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Sediments created through the processes of weathering can be moved from one place to another by a variety of physical agents, including ice, water and wind. This process is called erosion.
 - Different parts of the landscape are characterized by different sediments because the processes taking place there contribute to specific sediment types.
 - Clastic sedimentary rocks are formed from grains that are cemented together. The characteristics of the grains in each rock relate to the conditions under which the sediments were first formed, and what they have experienced since.
- AC SIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.
- AC SIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.
- AC SIS146 Reflect on scientific investigations including evaluating the quality of the data collected, and identifying improvements.
- AC SIS234 Use scientific knowledge and findings from investigations to evaluate claims based on evidence.
- AC SIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- explain how sedimentation events can lead to different patterns in sedimentary rocks.

Key vocabulary:

Sediment, graded bedding.

Equipment list

Each **GROUP** will require:

- 1.25 litre PET bottle with lid
- pebbles about 10 mm in size
- granules about 4 mm in size
- sand about 1 mm in size

Each **STUDENT** will require:

- **Notebook**



Things to consider

Ensure the pebbles can fit through the opening of the PET bottle.

To avoid spillage and mess in the classroom you might consider running this activity outside.

Provide basins for students to empty used sediments into. Ensure no sediments are tipped down sinks.

Teacher content information:

Please note that the stratigraphic convention is to describe the oldest event followed by subsequent events. So in this Activity students should start their descriptions noting what is happening at the bottom of the sequence, and then upward through the sequence from there. The deposition of sediments with different characteristics reflects different sedimentary environments. Characterising out a sequence of sediments that has been deposited over time allows us to reconstruct a geological history for a particular place.

If sediments are deposited by moving water (or wind), it typically takes a more energetic system to deposit larger particles. For example, conglomerates are typically more characteristic of a high energy river channel deposit, sandy sediments might be more characteristic of a river bank deposit and muddy sediments more characteristic of a flood plain deposit. Again this provides information on potential sedimentary environments for the sedimentary sequence we are studying.

If sediments are deposited into a column of water then the larger (or denser) particles will settle first followed by progressively smaller (less dense) particles. The sequence that is preserved is called graded bedding. Recognising this sequence in natural sediments allows you to identify which way-up the sedimentary beds were oriented when the sediments were deposited.

Lesson plan

- Step 1:** Recap with students the different sedimentary environments described in **Activity 4.2**. The movement of soil and sediments from one place to another is called erosion and is facilitated by wind, water or ice.
- Step 2:** Invite students to begin **Activity 1 Sediments**. Encourage students to note any observations in their **Notebooks**, record their sediments as a drawing or photo and answer the discussion questions.
- Step 3:** Discuss with students their findings and whether their sediments conformed with *Law 1: The oldest sediments are on the bottom (Nicholas Steno's Law of Superposition)* and *Law 2: The sediments are deposited horizontally (Nicholas Steno's Law of Original Horizontality)*.
- Step 4:** Invite students to begin **Activity 2 Graded bedding**. Encourage students to note any observations in their **Notebooks**, record their sediments as a drawing or photo and answer the discussion questions. Discuss students' findings.



6.3 How do fossils form?

Curriculum links

ACSSU153	Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales. KEY IDEAS Fossils are the remains of long dead organisms that have died and become preserved in sediments that have transformed into rock.
ACSHE 134	Scientific knowledge has changed peoples' understanding of the world and is refined as new evidence becomes available.
ACSIS146	Reflect on scientific investigations including evaluating the quality of the data collected, and identifying improvements.
ACSIS234	Use scientific knowledge and findings from investigations to evaluate claims based on evidence.
ACSIS148	Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- explain some of the processes and conditions required for the formation of fossils.

Key vocabulary:

Fossil, trace fossil, extinct, trilobite, palaeontologist, Naracoorte Caves, pitfall trap, megafauna.

Equipment list

Each STUDENT will require:

- Notebook**
- Internet access

Things to consider

Your school may have a fossil collection. Consider setting up a display of assorted fossils.

There may be an outcrop of shale or limestone nearby that contains fossils. Consider an excursion to such a site.

Teacher content information:

Trace fossils are fossils of the footprints, scratching or burrows left behind by an animal, and not the actual preserved animal remains.

Australia has a wealth of natural heritage in its various fossils sites, some of which are listed here - <https://australianmuseum.net.au/fossil-sites-of-australia>

Trilobites were hard-shelled arthropods that existed over 520 million years ago in Earth's ancient seas. They ranged in size from less than 1 mm to over 70 cm long. Most were predators, scavengers or detritivores.



Lesson plan

- Step 1:** What do your students already know about fossils? What environments favour fossil formation? Introduce your school fossil collection, if available.
- Step 2:** Invite students to complete the digital interactive that illustrates the sequence of fossil formation.
- Step 3:** Direct students to the **Notebook** and *Find out more* tasks which showcase Australia's heritage of fossil sites.



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⚠ Make considered choices selecting and using resources to minimise waste.

6.4 Is this the same as that?

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Fossils are the remains of long dead organisms that have died and become preserved in sediments that have transformed into rock.
 - Fossils provide valuable markers for identifying related rock strata occurring in different localities, even where their relationship is not evident at the Earth's surface.
- ACSIS146 Reflect on scientific investigations including evaluating the quality of the data collected, and identifying improvements.
- ACSIS234 Use scientific knowledge and findings from investigations to evaluate claims based on evidence.
- ACSIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- explain how the distribution of fossils in rock strata can be used to provide an indication of their relative ages.

Key vocabulary:

Stratigraphy, stratigraphic correlation, relative dating.

Equipment list

Each GROUP will require:

- **Activity sheet 6.4 Stratigraphy** (download from *Student Guide* or *Teacher Guide*)
- scissors

Each STUDENT will require:

- **e-Notebook** (download from *Student Guide*)

The CLASS will require:

- Activity sheet 6.4 Fossil cards answer (download from *Teacher Guide*)

Do not show students **Activity sheet 6.4 Fossil cards answer** until they have completed their sequences of the fossil cards.

Teacher content information:

Teaching about Earth's history is a challenge for all teachers. The idea of millions and billions of years is difficult for children and adults to comprehend. However, "relative" dating of time can be an easy concept for students to learn.



The use of fossils to identify different rock strata pre-dates the Theory of Evolution and a sound understanding of what fossils actually are. In a sense this simulation provides a very good impression of how fossils were used by the early 19th century to identify and map rock strata across the world. It didn't matter that some fossils were not accurately identified. As long as a pattern was observable then valid conclusions could be drawn about rock strata found in different localities. This included information about their relative ages. Of course this approach does not work for igneous rocks, and it is dependent on the existence of fossilised organisms occurring in particular strata, which is not always the case.

By correlating fossils from various parts of the world, scientists are able to give relative ages to particular strata. This is called relative dating. Relative dating tells scientists if a rock layer is "older" or "younger" than another. This would also mean that fossils found in the deepest layer of rocks in an area would represent the oldest forms of life in that particular rock formation. In reading Earth history, these layers would be "read" from bottom to top or oldest to most recent. If certain fossils are typically found only in a certain rock unit and are found in many places worldwide, they may be useful as index or guide fossils in finding the age of undated strata.

Key Idea:

Fossils provide valuable markers for identifying related rock strata occurring in different localities, even where their relationship is not evident at the Earth's surface.

Task 1 – Letter Cards – Answer

DM
OXD
NO
BN
UBN
AU
CA
TC

Task 2 – Fossil Cards – Answer

Refer to **Activity sheet 6.4 Fossil cards answer** (download from *Teacher Guide*)

Lesson plan

Task 1 – Letter Cards

- Step1:** Working in small groups, cut out the eight letter cards and spread them on the desk in front of you. Each letter on these cards represents a different fossil.
- Step 2:** The first (bottom) card in the sequence is TC. Hence the letters "T" and "C" represent fossils in the oldest rock layer. They are the oldest fossils, or the first fossils formed in the past for this sequence of rock layers.
- Step 3:** Sequence the eight cards by comparing letters that are common to individual cards, and therefore overlap. Look for a card that has either a "T" or "C" fossil on it. Since this card has a common fossil with the first card, it must go on top of the TC card. The fossils represented by the letters on this card are "younger" than the "T" or "C" fossils on the TC card.



Step 4: Sequence the remaining cards by the same process. When done you should have a vertical stack of cards with the top card representing the youngest fossils of this rock sequence and the TC card at the bottom of the stack indicating the oldest fossils.

Step 5: Download the **e-Notebook** for **Activity 6.4**. Copy the sequence of letter cards with the TC card at the bottom of the stack into your **e-Notebook** and answer questions 1-3.

Task 2 – Fossil Cards

Step 1: Cut out the eight Fossil Cards. Each card represents a particular rock layer with a collection of fossils that are found in that particular rock stratum. All of the fossils represented would be found in sedimentary rocks of marine origin.

Step 2: The oldest rock layer is marked with the letter “M”. Put it at the bottom of the stack.

Step 3: Look for a rock layer that has at least one of the fossils you found in the rock layer “M”. This rock layer would be younger as indicated by the appearance of new fossils in the rock stratum. Keep in mind that extinction is forever. Once an organism disappears from the sequence it cannot reappear later. Use this information to sequence the cards into a vertical stack of rock strata (stratigraphic column). Arrange them from oldest to youngest, with the oldest layer on the bottom.

Step 4: When you have completed your stack, check your stratigraphic column with the columns developed by other students. Do you agree? If not, discuss where your version is different and try to agree on a structure. Compare your stratigraphic column with **Activity sheet 6.4 Fossil cards answer**.

Step 5: Answer discussion questions 4 - 5 in your **e-Notebook**.

Extension:

1. Research the different fossils and identify where they are found on the geologic time scale. Does this information match the pattern in this activity?
2. Use this information to postulate possible ages for the different rock strata in this activity.

Source: Utah LessonPlans (<http://www.uen.org/Lessonplan/preview.cgi?LPid=16319>)



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6.5 Relative geological time

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Fossils provide valuable markers for identifying related rock strata occurring in different localities, even where their relationship is not evident at the Earth's surface.
- ACSIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- explain how the geological time scale was created and how it is used.
- describe how the appearance of different organisms in different rock strata relates to their evolutionary history.

Key vocabulary:

Relative geological time scale, radiometric dating, radioactive elements.

Equipment list

Each STUDENT will require:

- Notebook**
- internet access

Things to consider

Students will need access to the Geological Time Scale on page 55 of the *Student Guide*. In order to locate more information on the possible fossils found in each layer provide suitable texts or access to the internet.

Teacher content information:

On page 55 of the *Student Guide* students can see 1 km of sediments that have been deposited over a 100 million year period. This means that an average of only 1 mm of sediment is deposited per century.

Scientists have good evidence that Earth is very old; approximately four and one-half billion years. Scientific measurements such as radiometric dating use the natural radioactivity of certain elements found in rocks to help determine their age. Scientists also use direct evidence from observations of the rock layers themselves to find the relative age of rock layers. Specific rock formations are indicative of a particular type of environment existing when the rock was being formed. For example, most limestone represents marine environments, whereas, sandstones with ripple marks might indicate a shoreline habitat or riverbed.



The study and comparison of exposed rock layers or strata in different areas of Earth led scientists in the early 19th century to propose that the rock layers could be correlated from place to place. Locally, physical characteristics of rocks can be compared and correlated. On a larger scale, even between continents, fossil evidence can help in matching rock layers.

The Law of Superposition, which states that in an undisturbed horizontal sequence of rocks the oldest rock layers will be on the bottom, with successively younger rocks on top of these, helps geologists correlate rock layers around the world. This also means that fossils found in the lowest levels in a sequence of layered rocks represent the oldest record of life there. By matching partial sequences, the truly oldest layers with fossils can be worked out.

By correlating fossils from various parts of the world, scientists are able to give relative ages to particular strata. This is called relative dating. Relative dating tells scientists if a rock layer is “older” or “younger” than another. This would also mean that fossils found in the deepest layer of rocks in an area would represent the oldest forms of life in that particular rock formation. In reading Earth history, these layers would be “read” from bottom to top or oldest to most recent. If certain fossils are typically found only in a certain rock unit and are found in many places worldwide, they may be useful as index or guide fossils in finding the age of undated strata. By using this information from rock formations in various parts of the world and correlating the studies, scientists have been able to construct the geologic time scale: This relative geological time scale divides the vast amount of Earth history into various sections based on geological events (sea encroachments, mountain-building, and depositional events), and notable biological events (appearance, relative abundance, or extinction of certain life forms).

Lesson plan

Step 1: Discuss with students the slow rate of sediment deposition on page 54 of the *Student Guide*.

Step 2: Show students the Geological Time Scale on page 55 of the *Student Guide*. The fossils found in each layer provide a record of the evolutionary history of life on Earth.

Step 3: As a class explore the videos of the Grand Canyon. Students may begin the first of the Notebook tasks and describe how the Grand Canyon was carved out.

Step 4: The next **Notebook** task is ‘The Grand Canyon’s Rock Layers’. Students gain familiarity with geological periods and the corresponding fossils as they complete the table of Grand Canyon Rock Layers. Encourage students to search from textbooks or the internet for fossils from the various periods. Discuss with students the answers they provided to the **Notebook** questions.



6.6 How long does it take?

Curriculum links

- ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.
KEY IDEAS
- Although most rocks appear to be endlessly durable they have a life story. Rocks are broken down and rebuilt over very long periods of time.
 - Rock materials are not destroyed but cycle through the processes of formation of sedimentary, igneous and metamorphic rocks, over millions of years.
- ACSIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.
- ACSIS146 Reflect on scientific investigations including evaluating the quality of the data collected, and identifying improvements.
- ACSIS234 Use scientific knowledge and findings from investigations to evaluate claims based on evidence.
- ACSIS148 Communicate ideas, findings and evidence based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- describe examples of processes that take long periods of time in the rock cycle
- explain that different rocks in different environments might take different paths and different time periods through the rock cycle
- describe how the study of geology enables us to predict natural disasters and how to read rocks and sediments to see the history of the area.

Key vocabulary:

Epoch, period, era, eon, volcanologist.

Equipment list

Each CLASS will require:

- *Rock Cycle Game* materials. This version of the *Rock Cycle Game* comes from the Arkansas Geological Survey. It is available at <https://www.geology.arkansas.gov/docs/pdf/education/RockCycleGame.pdf>

Each STUDENT will require:

- **Notebook**
- Scissors, sticky tape
- internet access

Things to consider

Prepare the *Rock Cycle Game* kit in advance. Print the dice cutouts on card (if you make the dice out of stiff card and store them in a postage box these will be reusable from year to year). Laminate the station cards so that you can reuse the set over and over.



Staff can have access to a *Rock Cycle* diagram to refer to when assisting students with this activity.

Do not play the *Rock Cycle Game* for more than about 20 minutes or students will start losing interest in the game; especially if they are stuck in a Sedimentary Basin or similar setting for an extended period of time. There needs to be a reasonable time allowance after the activity to debrief about how long some of the geological processes took during the *Rock Cycle Game* (some processes take millions of years). Also some people cycle through the processes fairly quickly while others get stuck in a particular process, illustrating different potential pathways for the formation of different rock types. Finish this discussion by revisiting the *Rock Cycle* diagram and ensuring that students have a clear understanding of the processes that are summarised in this image.

Teacher content information:

The concepts in the Rock Cycle will be reinforced in the *Rock Cycle Game*. This is one of the few activities that links the processes of the rock cycle with the concept of geological (deep) time.

Key Idea:

Rock materials are not destroyed but cycle through the processes of formation of sedimentary, igneous and metamorphic rocks, over millions of years.

Lesson plan

Step 1: Follow the instructions on page 57 of the *Student Guide* for playing the *Rock Cycle Game*. When students have drawn their rock cycle diagrams and plotted the pathway their rock took, provide sufficient time for students to compare their results with others.

Step 2: Show the video and discuss how understanding geology can help us predict and understand natural disasters.



6.7 Forming landforms

Curriculum links

ACSSU153 Sedimentary, igneous and metamorphic rocks contain minerals and are formed by processes that occur within Earth over a variety of timescales.

KEY IDEAS

- Different environmental processes lead to the formation of very different landforms that can be observed in Australia today.

Lesson outcomes

At the end of this activity students will be able to:

- identify that different processes have produced different landforms in Australia

Key vocabulary:

Weathering, erosion, deposition, Uluru, glaciers, Blue Lake, Twelve Apostles.

Equipment list

Each STUDENT will require:

- **Notebook**
- internet access

Things to consider

This activity brings together a range of ideas covered throughout the unit. It would be useful to include some whole class discussion to draw out these ideas. This could also provide an opportunity for the teacher to gauge how well students have mastered many of these ideas.

Teacher content information:

Commonly the underlying geology influences the geomorphic landforms that result. The degree of weathering and erosion of the landscape also influences the geomorphology. If we are aware what geology underlies different landforms, and we understand the weathering and erosion processes taking place in a landscape, we can make evaluations about the types of soils that will form in different parts of the landscape. This, in turn influences how water moves through the landscape, and why different vegetation types thrive in different parts of the landscape.

Key Idea:

Different environmental processes lead to the formation of very different landforms that can be observed in Australia today.

Lesson plan

Step 1: Discuss with students the pictures of Australia's iconic landforms on page 59 of the *Student Guide*. Ask students to match the descriptions with each photo.

Step 2: Invite students to click on the digital link and explore each landform in greater detail. The **Notebook** tasks require students to summarise their findings of how geological processes contributed to these landforms.



6.8 Sample test

A sample summative test (editable version) and marking scheme are available to teachers by email request from *Science by Doing* (sbd@science.org.au). In your email request please include your school (or university) and contact details.

Note - *Science by Doing* provides sample assessment items and whilst every effort has been made, the security of these items cannot be guaranteed. *Science by Doing* encourages teachers to modify the items to suit individual teaching programs.



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Click [here](#) for a full version of unit acknowledgements and sources or logon to www.sciencebydoing.edu.au.

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